Imp property (Linear Trans)

① Origin (0) → must be fixed

② All lines must remain lines

---

16 Magnitude and Unit vector

Vector length

$\vec{x} \in R^n$

↳ belong to Real no. (n-dimension)

vector

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

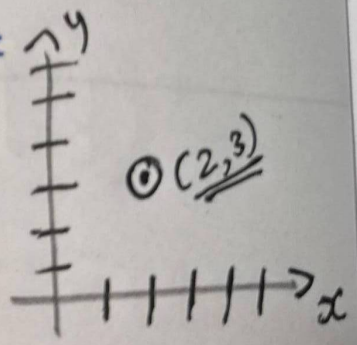
Unit vector

↳ vector has a length of 1

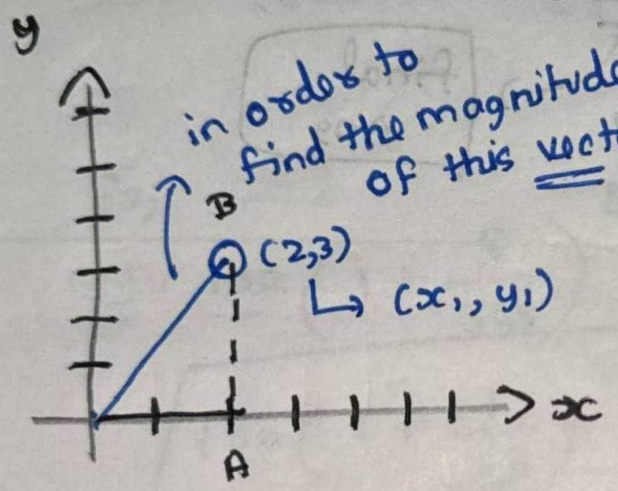
$$A = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

→ x

↳ y



Continuation



in order to find the magnitude of this vector

$$\vec{A} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \in R^2$$

$x \to 2$, $y \to 3$

$$OB = \|\vec{A}\| = \sqrt{(OA)^2 + (OB)^2}$$

$$= \sqrt{x_1^2 + y_1^2}$$

$$\|\vec{A}\| = \sqrt{4+9}$$

$$= \sqrt{13}$$

(if it is 3 dimension)

$$\vec{B} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$||\vec{B}|| = \sqrt{x_1^2 + x_2^2 + x_3^2}$$

So, in order to find the vector length

$$\boxed{||\vec{x}|| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}}$$

Unit Vector

$$||\vec{v}|| = 1 \longrightarrow \text{represented by } \hat{u}$$

$$\vec{V} = \begin{bmatrix} V_1 \\ V_2 \\ \vdots \\ V_n \end{bmatrix} \longrightarrow \vec{u} \quad \text{where } ||\vec{v}|| = 1$$

we have to convert this to

magnitude

Now,

$$\boxed{\vec{u} = \frac{1}{||\vec{v}||} \cdot \vec{v}}$$

In this context it is called Scalar multiplication

DOT PROD
↳ in general

Example

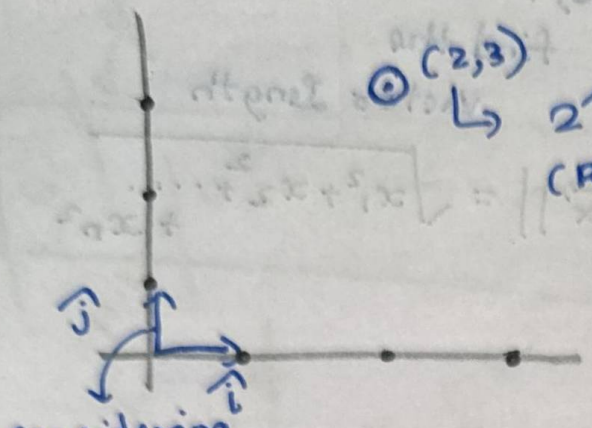
$$\vec{v} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \in R^3 \qquad ||\vec{v}|| = \sqrt{(1)^2 + (2)^2 + 0} = \sqrt{5}$$

So,

$$\vec{u} = \frac{1}{\sqrt{5}} \cdot \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \\ 0 \end{bmatrix}$$

Now,

$$||\vec{v}|| = \sqrt{(1/\sqrt{5})^2 + (2/\sqrt{5})^2 + 0^2} \longrightarrow \sqrt{1/5 + 4/5 + 0} \rightarrow \sqrt{5/5} \rightarrow \sqrt{1} = \boxed{1}$$



$\odot$ (2,3)

↳ $2\hat{i} + 3\hat{j}$

(Representation of that vector)

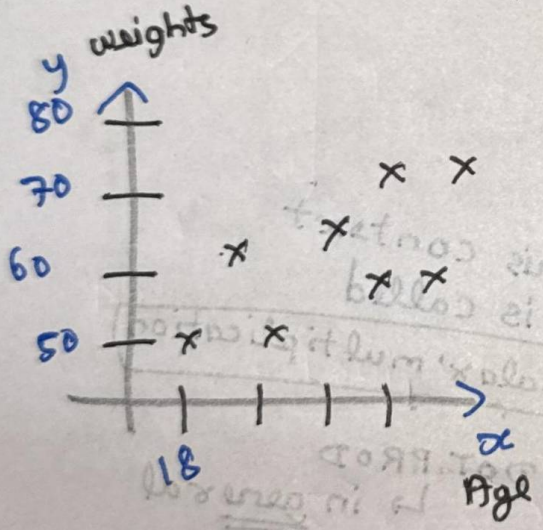
here $\hat{i}, \hat{j}$ are called **unit vectors**

considering this as my **unit vector**

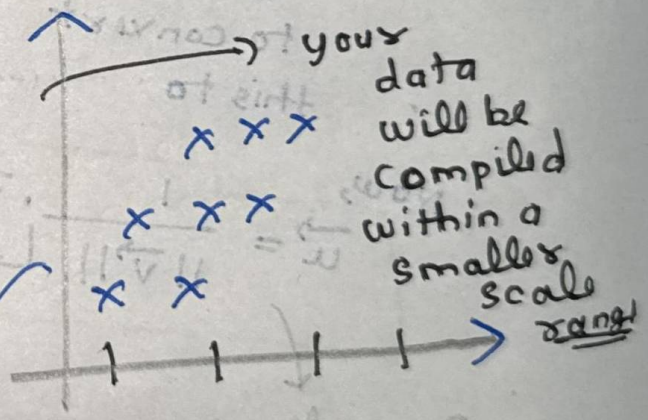
(Real time Scenario)

Normalization

whatever can be the vector size → if you want to find out a unit vector (vector length = 1)



if you want to **normalize this** → your data will be compiled within a smaller scale range



(All the values will be converted in a range b/w (0 - 1))

This will help you to improve **optimisation process**